

Adult Income Census Dashboard

Power BI Project Documentation

Developed By: Kurli Lakshmieswar Reddy

Tools Used:

- Microsoft Power BI | Microsoft Excel | Power Query | DAX (Data Analysis Expressions)

DATE

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1. Project Overview

The **Adult Income Census Dashboard** is an interactive Business Intelligence dashboard developed using Microsoft Power BI. The objective of the project is to analyze demographic and income-related patterns using the Adult Census Income dataset.

The dashboard enables users to explore how education, occupation, age, gender, marital status, and working hours influence income levels. It includes dynamic KPIs, interactive slicers, and DAX-generated insights that update automatically based on user selections.

2. Business Problem

Organizations often need to understand which demographic factors influence income distribution. Static reports make it difficult to explore this information from different perspectives.

This dashboard was developed to provide an interactive solution where users can:

- Analyze income distribution
 - Compare high-income and low-income populations
 - Explore demographic trends
 - Generate business insights dynamically
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3. Dataset Information

Dataset Name

Adult Census Income Dataset

Records

Approximately 33,000 records

Important Columns

- Age
 - Education
 - Occupation
 - Gender
 - Relationship
 - Marital Status
 - Hours Worked Per Week
 - Capital Gain
 - Capital Loss
 - Native Country
 - Income Category
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4. Data Cleaning

The raw dataset was cleaned before visualization.

The cleaning process included:

- Removing unnecessary columns
- Handling missing values
- Standardizing categorical values
- Correcting inconsistent data

- Formatting numerical fields
- Preparing the data model for reporting

Excel was used for initial cleaning, while Power Query was used for additional transformations before loading the data into Power BI.

5. Data Modeling

The dashboard uses a structured data model within Power BI.

Relationships were created between the transformed tables where required, enabling efficient filtering and reporting.

6. Dashboard Components

KPIs

The dashboard contains dynamic KPI cards:

- Average Age
- Total Individuals
- High Income Percentage (>50K)
- Low Income Percentage (≤50K)

All KPIs automatically update when slicers are applied.

Interactive Slicers

Users can filter the dashboard using:

- Gender
- Relationship
- Occupation

Every chart, KPI, and insight updates automatically according to the selected filters.

Visualizations

The dashboard includes:

- Income Distribution by Education
- Income by Gender
- Marital Status vs Income
- Income by Hours Worked
- Top 5 Occupations by Population
- Age Group vs Income

These visualizations help identify income trends across multiple demographic factors.

7. Dynamic Insights

One of the major features of the dashboard is the Dynamic Insights Panel.

Instead of using static text, DAX measures generate insights that automatically change based on user selections.

Examples include:

- Percentage of individuals earning above and below \$50K
- Average education level
- Working hours trends
- Capital gain and capital loss averages
- Largest race group

This allows users to obtain meaningful observations instantly.

8. DAX Features Used

The dashboard makes use of DAX measures for:

- Dynamic KPIs
- Percentage calculations

- Average calculations
- Count calculations
- Dynamic business insight generation
- Filter-aware reporting

These measures respond automatically to slicer selections and filter context.

9. Key Insights

Some important observations from the dashboard include:

- Approximately 24% of individuals earn above \$50K.
 - Around 76% earn \$50K or less.
 - Higher education levels generally show a greater proportion of high-income individuals.
 - Working more than 40 hours per week is common among higher-income individuals.
 - Occupation significantly influences income distribution.
 - Income patterns vary across gender and marital status.
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10. Skills Demonstrated

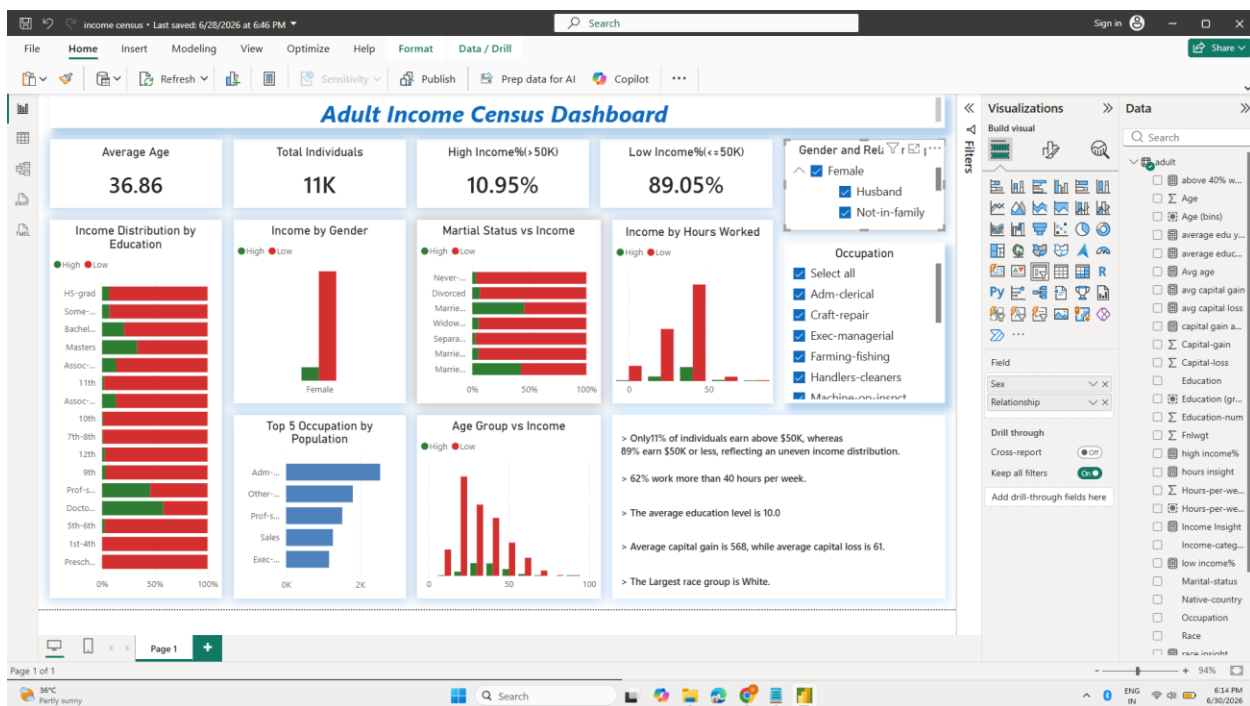
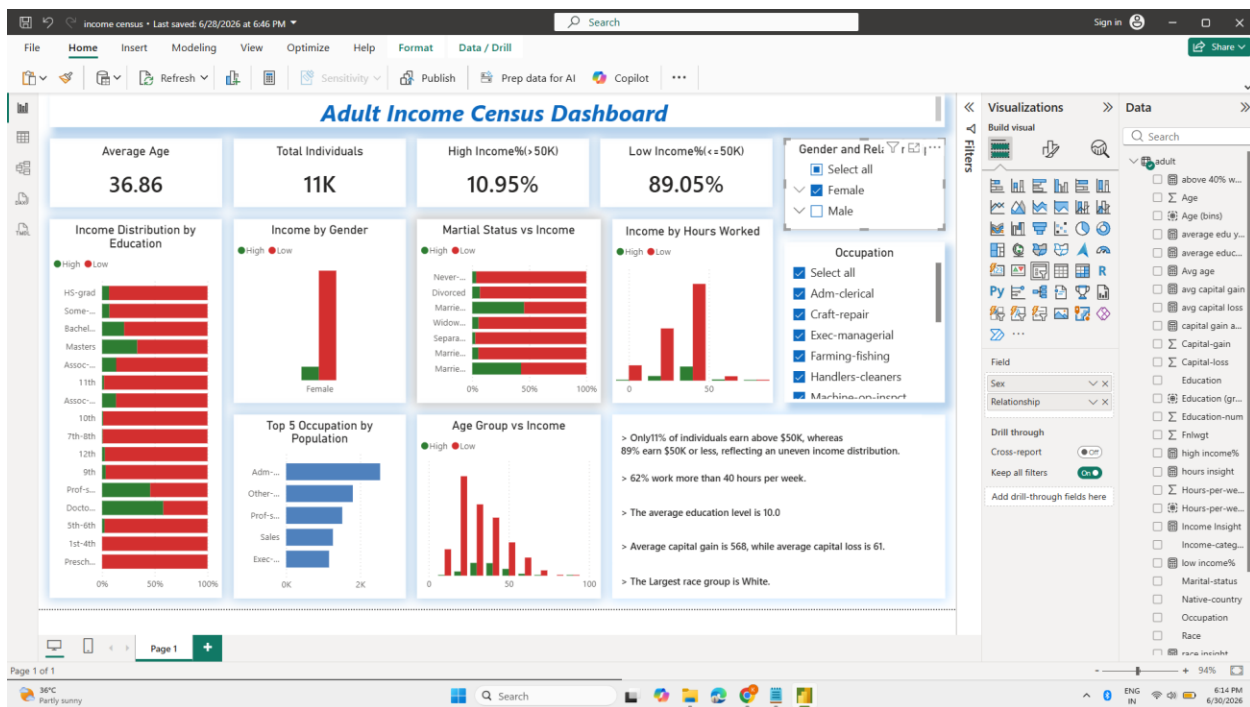
This project demonstrates practical knowledge of:

- Data Cleaning
- Data Transformation
- Power Query
- DAX
- Interactive Dashboards
- KPI Development
- Dynamic Insights
- Data Visualization

- Business Intelligence
- Dashboard Design

11. Dashboard Preview





12. Future Enhancements

Possible improvements include:

- Drill-through pages

- Custom tooltip pages
 - Time-series analysis
 - Additional demographic analysis
 - Advanced DAX calculations
 - Integration with SQL Server
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13. Conclusion

The Adult Income Census Dashboard demonstrates an end-to-end Business Intelligence workflow, beginning with raw data preparation and ending with an interactive reporting solution.

The project showcases practical skills in Excel, Power Query, DAX, and Power BI while providing meaningful business insights through interactive visualizations and dynamic reporting.

About the Author

Kurli Lakshmieswar Reddy

Skills

- SQL
- Power BI
- Microsoft Excel
- Power Query
- DAX
- Data Visualization
- Business Intelligence